

The glide axioms & transforms — one congruence, two roles, four laws

The whole family, put in order: one primitive — a line congruence with a **role** (edges or rays) — four axioms for the edge role, one transform (the inversion  $T$  that swaps the role), and everything else demoted to theorem. Two frames on the drawing — a **picture frame** that draws true-size and a **depth frame** that is its translated, stretched copy — one shared **station**, and a projective measure. The three pins use up all three Möbius freedoms, so the law  $\lambda = k/(k+1)$  is not chosen: it is forced. Drag anywhere — the two frames, the aperture plate, and the light test move as one object. The companion paper's single dial  $D$  is a path across this square ( $r_u = 1-k$ ,  $r_v = 1$ ), not a second axis.

general glideisotropicvanishing-axisone-pointpushbroomorthographic

Left = the two frames (drag a purple corner) · Middle = the aperture plate (drag the marker) · Right = the light test (drag to orbit) · arrow keys nudge

The axioms

|  |       |
|--|-------|
|  | Axiom |
|--|-------|

|   |                                                                                                                                                                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | <b>The primitive.</b> A line congruence from the catalog — the parallel bundle, a pencil, or a hyperbolic linear congruence (two skew focal lines: the directing axes) — together with a <b>role</b> : edges or rays. |
| P | The $z = 0$ copy is the identity: the picture plane draws true-size. A square is a square.                                                                                                                            |
| D | <b>Every</b> depth draws as an axis-aligned, translated, stretched copy of the picture grid — grid parallel to grid at all depths. The far copy is the <b>depth frame</b> .                                           |
| S | <b>One station.</b> A single depth, shared by both channels, maps beyond the frame.                                                                                                                                   |

|   |                                                                                                                                              |
|---|----------------------------------------------------------------------------------------------------------------------------------------------|
| M | Each channel reads depth projectively <b>as a whole</b> — scale and offset together: every channel coordinate is a Möbius function of depth. |
|---|----------------------------------------------------------------------------------------------------------------------------------------------|

### The theorems

|    |                                                                                                                 |
|----|-----------------------------------------------------------------------------------------------------------------|
|    | Theorem                                                                                                         |
| T1 | <b>The law.</b> $\lambda = k/(k+1)$ on every line, at every rate — no freedom remains after the pins.           |
| T2 | <b>Straight tracks.</b> The shared station keeps every depth-track a straight line in the picture.              |
| T3 | <b>Cameras carry no frame.</b> Split the station and the far image collapses to a point — the cross-slit frame. |
| T4 | <b>Two frames are the whole datum.</b> Picture frame + depth frame force every mark of the projection           |

|    |                                                                                                                                                    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| T5 | <b>The directing fates.</b> Merge into the limit ( $r = 1$ ) · survive finite ( $0 < r < 1$ ) · leave for infinity ( $r = 0$ ). point is the bend. |
| T6 | <b>The light theorem.</b> Fibers are straight at the four corners only — pinhole, orthographic, the two p is impenetrable.                         |
| T7 | <b>The armature identity.</b> The glide lines are the rays of the cross-slit congruence of the two directing cross-slit, the law is the glide.     |

|    |                                                                                                                                                                                                                                                  |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T8 | <b>The inversion is an involution.</b> T swaps the role on the same congruence, and $T^2 = \text{id}$ : each reading of a congruence, the congruence determines the other reading, and the round trip glide $\rightarrow$ camera $\rightarrow$ g |
| T9 | <b>The two fixed loci, strictly nested.</b> Drawn lines are T-fixed exactly on the degenerate catalog — the $r = 0$ edges (slit + $\infty$ ), the corners; full maps only at the corners. Fixmaps $\subsetneq$ Fixlines.                         |

### The directing fates

| Rate                                                                                                                         | The directing point            | That channel's ray                            | Name                 |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------|----------------------|
| $r = 0$                                                                                                                      | leaves for infinity            | straight                                      | orthographic channel |
| $0 < r < 1$                                                                                                                  | survives — finite and separate | <b>bent</b> — the directing point is the bend | glide channel        |
| $r = 1$                                                                                                                      | merges into the depth limit    | straight                                      | perspective channel  |
| <b>Light passes <math>\Leftrightarrow</math> no survivor on the line.</b> Both channels must give up their directing points: |                                |                                               |                      |

### Transformational logic — the moves

| Move                                  | On the square             | On the drawing                                                                                                                                                                               |
|---------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| stretch the depth                     | slide the marker          | the whole projection follows — the law is forced (T1, T4)                                                                                                                                    |
| merge a directing point (r)           | reach an $r = 1$ edge     | the frame collapses to a segment; that channel straightens                                                                                                                                   |
| release a directing                   | reach an $r = 0$ edge     | the frame weds the picture in that axis; that channel straightens                                                                                                                            |
| surrender or both                     | land on a corner          | <b>light passes:</b> pinhole · orthographic · the two pushbroom slits                                                                                                                        |
| swap the channel                      | reflect across the        | the drawing turns a quarter-turn (check_axioms 6)                                                                                                                                            |
| <b>transpose the role (T)</b>         | stay at the same point of | the same congruence re-read, edges $\Leftrightarrow$ rays: the shadow armature is shared always; the camera's tracks stay on the drawn lines only over the degenerate catalog: the full maps |
| split the station ( $S \rightarrow$ ) | <b>leave the square</b>   | the tracks curve, one law becomes two, the frame collapses to a point: the cross-slit camera (T2, T3)                                                                                        |
| keep the frame                        | —                         | the armature identity: the same lines carry either graduation (T7)                                                                                                                           |

### Outside anchors

| This system | Established name | Where |
|-------------|------------------|-------|
|-------------|------------------|-------|

|                                         |                                                                                                                          |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| the two-axis congruence; the directing  | hyperbolic linear congruence; its two focal lines                                                                        | classical line geometry (Plücker; Pottmann–Wallner, <i>Computational</i> |
| reading 2 on that congruence            | the Crossed-Slits (X-Slits) projection; perpendicular axis-parallel slits = POX-Slits                                    | Pajdla 2002; Zomet–Feldman–Peleg–Weinshall, <i>IEEE PAMI</i>             |
| the corner-and-edge camera              | General Linear Cameras (GLC)                                                                                             | Yu–McMillan, <i>ECCV</i> 2004                                            |
| cameras as line                         | “What is a Camera?”                                                                                                      | Ponce, <i>CVPR</i> 2009                                                  |
| two frames, linear interpolation, rank  | ray-bundle cross-sections $P' = (1-d)P + dI$                                                                             | Adams–Levoy, <i>General Linear Cameras with Finite Aperture</i>          |
| the convergent-solid ancestor           | relief perspective — a <b>central</b> perspective collineation (image plane + vanishing plane); reading 1's orthographic | classical descriptive geometry                                           |
| flattened solids under parallel viewing | the (generalized) bas-relief ambiguity                                                                                   | Belhumeur–Kriegman–Yuille, <i>IJCV</i> 1999                              |

Anchors are reconnaissance-level (a handful of searches), to be confirmed in a full literature pass. The transposition itself — one congruence, two roles, two nested fixed loci — was not found under any name.

Axiom S is the personality of the family. Keep the one station and you glide: the law is single, the tracks are straight, the depth frame survives, and the surface is closed to light everywhere but the corners. Split the station and you photograph: the law divides, the tracks curve, the frame collapses to a point, and a straight ray threads two slits. The square is the space of depth frames; the perimeter is where they degenerate; the corners are where light gets in. And over it all sits the inversion: the same congruence worn as edges or as rays — an involution whose drawn lines survive on the degenerate catalog and whose full maps agree only at the corners. The laws above are the edge role's half of that symmetry.

Verified in `check_axioms.js` (7: the pins force the law; the shared station gives straight tracks and splitting it curves them and destroys the frame; the directing trichotomy; light at the corners only; channel

swap = the diagonal reflection; translation never bends a ray) and check\_axiom\_gap.js (5: stating D for the far plane alone admits a rotating counterexample — true-size picture, affine far frame, rotated in between — so D is strengthened to every depth, which supplies the channels and forces the glide scale uniquely) and check\_dof\_audit.js (4: the repair itself was still gappy — the offsets were unpinned, and a wandering-offset projection passed P, D and scale-M while curving its tracks; auditing all four depth-functions and requiring M of the whole channel map forces the two-plane interpolation uniquely) and check\_transposition.js (6:  $T^2 = \text{id}$  — the round trip glide  $\rightarrow$  camera  $\rightarrow$  glide is the identity; the congruence is T's invariant; drawn lines are T-fixed exactly on the degenerate catalog; full maps only at the corners; the nesting is strict; the catalog is closed) — standing on check\_inversion (10), check\_two\_planes (8), check\_ray\_bend (3), check\_crossslit\_frame (9), check\_split (9), check\_vaxis (8), check\_isotropy (7) and check\_atlas (8). The page re-runs six checks on every drag. Audit total 317 · snapshot 2026-07 · matches atlas page.